Jacob A. Johnson

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EDUCATION

Ph.D. in Statistics Expected May 2029

University of Wisconsin-Madison, *Madison, WI*

B.S. in Statistical Science, Minor in Mathematics June 2024

Brigham Young University, *Provo, UT*

- *summa cum laude* (4.00/4.00 GPA)
- National Merit Scholar

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Statistics Intern May 2025 – Present

Sandia National Laboratories, *Albuquerque, NM*

- Add 1-2 bullet points describing your spatial statistics work and its impact on the ESTCP project goals

Financial Data Analytics Intern May 2024 – Aug 2024

Williams International, *Pontiac, MI*

- Created a statistical model to forecast inventory using historical production and usage patterns.
- Developed an anomaly detection program to monitor and identify deviations in expenses, detecting unusual spending patterns and potential fraud across business units.
- Automated forecasting and anomaly detection processes, reducing manual oversight and providing executives with actionable insights.

Research Assistant May 2022 – April 2024

Brigham Young University, *Provo, UT*

- Utilized Bayesian machine learning techniques to combine climate data products for predicting average monthly precipitation in the Himalayan Mountains.
- Developed new subsampling methods in R to speed up computation of Gaussian Process calculations for large spatial data.

PUBLICATIONS

- Heaton, M.J., **Johnson, J.A.** "Minibatch Markov Chain Monte Carlo Algorithms for Fitting Gaussian Processes." *Bayesian Analysis*. Advance Publication 1-23, 2024. https://doi.org/10.1214/24-BA1459.
- **Johnson, J.A.**, Heaton, M.J., Christensen, W.F. et al. "Fusing Climate Data Products

Using a Spatially Varying Autoencoder." *JABES* (2024). https://doi.org/10.1007/s13253-024-00657-3.

HONORS & AWARDS

- UW-Madison Statistics Department Outstanding Performance as a TA award 2024 – 2025
- ASA Section on Statistics and the Environment Student Paper Competition Winner 2024

PRESENTATIONS

Oral Presentations

- "Climate Data Melding via Spatially varying Bayesian Autoencoder." Joint Statistical Meetings (JSM), Toronto, ON. August 2023.
- "Climate Data Melding via Spatially varying Bayesian Autoencoder." CPMS Student Research Conference, Provo, UT. February 2023.

Poster Presentations

- "Climate Data Melding via Spatially varying Bayesian Autoencoder." ENVR Workshop, Provo, UT. October 2022.

TEACHING EXPERIENCE

Teaching Assistant, *University of Wisconsin-Madison* Fall 2024 – Present

- STAT 324 - Introductory Statistics for Engineers (Fall 2024, Spring 2025)
- STAT 240 - Data Science Modeling I (Fall 2025)

Teaching Assistant, *Brigham Young University* Sept 2021 – April 2022

- STAT 240 - Probability and Inference I (Fall 2021, Spring 2022)

TECHNICAL SKILLS

- **Languages & Tools:** R, Python, SQL, Linux shell processes.